Amrita School of Artificial Intelligence, Faridabad
Amrita Vishwa Vidyapeetham

Course Code: **23MAT214**

Mathematics for Intelligent System-4

Part- 1

Special Matrices: Fourier Transformation

BTech 4th Sem

Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

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Why Do We Transform Data?

The vector

$$\mathbf{x} = \begin{bmatrix} 2 \\ 4 \\ 1 \\ 0 \end{bmatrix}$$

shows the values directly.

But it does **not** easily answer questions such as:

- Does the data repeat?
- Is there a regular pattern?
- Is the data smooth or oscillatory?

To understand these properties, we use a **different representation** of the same data.

AMRITA VISHWA VIDYAPEETHAM
Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

Continuous Fourier Transform (Background)

For a continuous function $f(t)$,the Fourier Transform is defined as:

$$F(\omega) = \int_{-\infty}^{\infty} f\,(t)\, e^{-i\omega t}\, dt$$

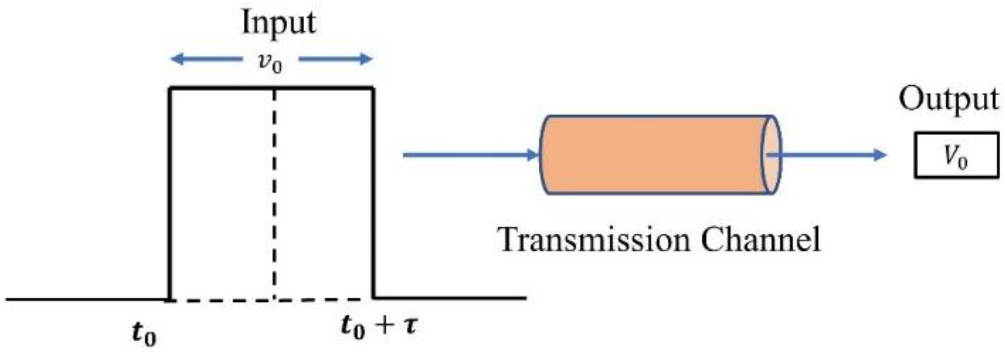
This formula:

- Is important in theory
- Comes from mathematical analysis
- Cannot be computed directly by computers (infinite limits)

Therefore, we use a **discrete version** for computation.

AMRITA VISHWA VIDYAPEETHAM DEEMED TO BE UNIVERSITY
Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

Fourier Transform of unit Signal



$$F_0(\omega,\tau) = \int_{-\infty}^{\infty} v_0 e^{-i\omega t}\,dt = \int_{t_0}^{t_0+\tau} v_0 e^{-i\omega t}\,dt$$

$$= \frac{v_0}{i\omega} e^{-i\omega t_0}\left(1 - e^{-i\omega\tau}\right)$$

AMRITA VISHWA VIDYAPEETHAM DEEMED TO BE UNIVERSITY
Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

Why matrices for Fourier Transform?

Fourier Transform converts a signal from time domain to frequency domain. In continuous form,

$$X(\omega) = \int_{-\infty}^{\infty} x\,(t) e^{-i\omega t}\, dt$$

But in computers, we work with **finite-dimensional vectors**, not functions. So Fourier Transform becomes a **linear transformation**:

$$\mathbf{X} = \mathbf{F}\mathbf{x}$$

where

$\mathbf{x} \in \mathbb{C}^N$ is a signal vector

$\mathbf{F} \in \mathbb{C}^{N \times N}$ is a **special matrix**

$\mathbf{X} \in \mathbb{C}^N$ is the frequency-domain vector

So Fourier Transform is **just matrix multiplication**.

AMRITA VISHWA VIDYAPEETHAM DEEMED TO BE UNIVERSITY
Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

Discrete Fourier Transform

For a discrete signal $x[n]$,$n = 0,1, \dots, N-1$,

$$X[k] = \sum_{n=0}^{N-1} x\,[n] e^{-i\frac{2\pi}{N}kn}$$

Inverse:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X\,[k] e^{i\frac{2\pi}{N}kn}$$

Key observation:

This is **linear in** $x[n]$ →hence representable by a matrix.

AMRITA VISHWA VIDYAPEETHAM DEEMED TO BE UNIVERSITY
Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

Fourier Transform as a Linear Operator

- A Fourier Transform is fundamentally a **linear transformation**.

  For any two signals x_1, x_2 and scalars a, b,

$$\mathcal{F}[ax_1 + bx_2] = a\mathcal{F}[x_1] + b\mathcal{F}[x_2]$$

- Any linear transformation between finite-dimensional vector spaces can be represented by a **matrix**.
- Therefore, the Discrete Fourier Transform (DFT) can be written as a matrix multiplication.

AMRITA VISHWA VIDYAPEETHAM
Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

Discrete Fourier Transform Matrix

Define the primitive root of unity:

$$\omega = e^{-i\frac{2\pi}{N}}$$

Then DFT becomes:

$$X[k] = \sum_{n=0}^{N-1} x\,[n]\omega^{kn}$$

For a discrete signal $x[n]$,$n = 0,1, \dots, N-1$,

$$X[k] = \sum_{n=0}^{N-1} x\,[n]e^{-i\frac{2\pi}{N}kn}$$

Write this in matrix form:

$$\begin{bmatrix} X[0] \\ X[1] \\ \vdots \\ X[N-1] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \omega & \omega^2 & \cdots & \omega^{N-1} \\ 1 & \omega^2 & \omega^4 & \cdots & \omega^{2(N-1)} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & \omega^{N-1} & \omega^{2(N-1)} & \cdots & \omega^{(N-1)^2} \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix}$$

This matrix is called the **DFT matrix**, usually denoted:

$$\mathbf{F}_N = \left[\omega^{kn}\right]_{k,n=0}^{N-1}$$

This is the **central special matrix** of Fourier theory.

AMRITA VISHWA VIDYAPEETHAM
Dr. Ahirdan Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

Discrete Fourier Transform (DFT)

Let a discrete-time signal be represented as a vector:

$$\mathbf{x} = \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix} \in \mathbb{C}^N$$

The DFT of $\mathbf{x}$is defined as:

$$X[k] = \sum_{n=0}^{N-1} x\,[n] e^{-i\frac{2\pi}{N}kn}, k = 0,1, \dots, N-1$$

Define the complex constant:

$$\omega = e^{-i\frac{2\pi}{N}}$$

Then,

$$X[k] = \sum_{n=0}^{N-1} x\,[n] \omega^{kn}$$

AMRITA VISHWA VIDYAPEETHAM DEEMED TO BE UNIVERSITY
Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

Fourier (DFT) Matrix

The DFT can be written in matrix form:

$$\mathbf{X} = \mathbf{F}_N \mathbf{x}$$

where the **Fourier matrix** $\mathbf{F}_N \in \mathbb{C}^{N \times N}$ is defined as:

$$\mathbf{F}_N = \begin{bmatrix} \omega^{0\cdot 0} & \omega^{0\cdot 1} & \cdots & \omega^{0\cdot (N-1)} \\ \omega^{1\cdot 0} & \omega^{1\cdot 1} & \cdots & \omega^{1\cdot (N-1)} \\ \vdots & \vdots & & \vdots \\ \omega^{(N-1)\cdot 0} & \omega^{(N-1)\cdot 1} & \cdots & \omega^{(N-1)(N-1)} \end{bmatrix}$$

Compactly:

$$\left(\mathbf{F}_N\right)_{k,n} = \omega^{kn}$$

AMRITA VISHWA VIDYAPEETHAM DEEMED TO BE UNIVERSITY
Dr. Amitavan Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

Real-life example: Mixing colors of light (very intuitive)

Imagine this situation you can say in class:

You have **4 lamps** in a room (red, green, blue, violet) .
Each lamp emits light at a **fixed time instant**.
What you *see* with your eyes is a **mixture of different colors** (frequencies).

Here:

- **Time domain** → how bright each lamp is
- **Frequency domain** → how much of each color (red, green, blue, violet) is present

AMRITA VISHWA VIDYAPEETHAM
Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

Step 1: Discrete signal as “brightness readings”

Suppose we observe brightness at **4 equally spaced time instants**:

$$x[0], x[1], x[2], x[3]$$

Think of:

- $x[0]$:brightness at time 0
- $x[1]$:brightness at time 1
- $x[2]$:brightness at time 2
- $x[3]$:brightness at time 3

So the signal vector is:

$$\mathbf{x} = \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix}$$

This is **what we measure directly**.

AMRITA VISHWA VIDYAPEETHAM DEEMED TO BE UNIVERSITY
Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

Could we take N = 8, 16, 1000?

Yes — absolutely.

N is a modeling choice, not a physical property.

Real system	N choice	Meaning
Room light	N = 4	4 measurements
ECG signal	N = 1000	1000 samples
Audio clip	N = 44100	1 second audio
Image row	N = 512	512 pixels

AMRITA VISHWA VIDYAPEETHAM DEEMED TO BE UNIVERSITY
Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

Step 2: What Fourier Transform really asks

Fourier Transform asks:

"How much of each **pure oscillation** is hidden inside this signal?"

For $N = 4$, there are **4 basic oscillations**:

- Constant (DC)
- Slow oscillation
- Faster oscillation
- Fastest oscillation

Each oscillation is a **complex sinusoid**.

AMRITA VISHWA VIDYAPEETHAM
Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

Step 3: Primitive root of unity (key intuition)

$$\omega = e^{-i\frac{2\pi}{N}}$$

We define:

$$\omega = e^{-i\frac{2\pi}{4}} = e^{-i\frac{\pi}{2}}$$

Numerically:

$$\omega = \cos\left(\frac{\pi}{2}\right) - i\sin\left(\frac{\pi}{2}\right) = -i$$

So powers of ω cycle:

$$\omega^0 = 1, \omega^1 = -i, \omega^2 = -1, \omega^3 = i$$

This cycling behavior is **why Fourier basis captures periodicity**.

AMRITA VISHWA VIDYAPEETHAM DEEMED TO BE UNIVERSITY
Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

Step 4: Meaning of the DFT formula (core intuition)

Your equation:

$$X[k] = \sum_{n=0}^{N-1} x\,[n]\,\omega^{kn}$$

In words:

To find the amount of the **k-th frequency**,
multiply the signal by the **k-th oscillation pattern**,
then **add everything**.

So:

- $x[n]$ →brightness at time n
- ω^{kn} →rotating complex wave
- Summation → "how much do they match?"

This is **correlation with a wave**.

AMRITA VISHWA VIDYAPEETHAM
Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

Step 5: Matrix form — what it *means* physically

For $N = 4$, the DFT matrix is:

$$\mathbf{F}_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & \omega & \omega^2 & \omega^3 \\ 1 & \omega^2 & \omega^4 & \omega^6 \\ 1 & \omega^3 & \omega^6 & \omega^9 \end{bmatrix}$$

Each **row** is:

one frequency pattern sampled at 4 time points

Each **column** corresponds to:

one time instant

AMRITA VISHWA VIDYAPEETHAM
Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

Step 6: Matrix multiplication = weighted mixing

$$\begin{bmatrix} X[0] \\ X[1] \\ X[2] \\ X[3] \end{bmatrix} = \mathbf{F}_4 \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix}$$

Interpretation:

- Row 1: adds all samples → **average brightness**
- Row 2: compares signal with slow oscillation
- Row 3: compares with medium oscillation
- Row 4: compares with fast oscillation

So each $X[k]$answers:

"How strongly does this signal contain the k-th oscillation?"

AMRITA VISHWA VIDYAPEETHAM DEEMED TO BE UNIVERSITY
Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

Step 7: Why linearity ⇒ matrix (crucial sentence for students)

Your key observation:

This is linear in $x[n]$

Explain it like this:

If we change the signal slightly:

- doubling brightness → doubles output
- adding two signals → outputs add

That means:

$$\mathbf{X} = \mathbf{F}_N \mathbf{x}$$

is **just a linear change of coordinates**.

Time domain → Frequency domain

is **not magic**, just a **basis change**.

AMRITA VISHWA VIDYAPEETHAM DEEMED TO BE UNIVERSITY
Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

Step 8: Inverse transform — reconstructing the signal

Inverse formula:

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X\,[k]e^{i\frac{2\pi}{N}kn}$$

Meaning:
Take all frequencies,
rotate them back,
add them,
and divide by N
So:

- Forward FT → decompose signal
- Inverse FT → rebuild signal

Like:
Breaking a color into RGB
and then recombining RGB back to the same color

AMRITA VISHWA VIDYAPEETHAM DEEMED TO BE UNIVERSITY
Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

Common setup If N=4

Let

$$N = 4, \omega = e^{-i\frac{2\pi}{4}} = -i$$

DFT matrix:

$$\mathbf{F}_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix}$$

DFT formula:

$$\mathbf{X} = \mathbf{F}_4\mathbf{x}$$

AMRITA VISHWA VIDYAPEETHAM DEEMED TO BE UNIVERSITY
Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad

Real-life: Constant room temperature

Suppose temperature is constant at all times:

$$x[0] = 5,\ x[1] = 5,\ x[2] = 5,\ x[3] = 5$$

So,

$$\mathbf{x} = \begin{bmatrix} 5 \\ 5 \\ 5 \\ 5 \end{bmatrix}$$

Now compute:

$$\mathbf{X} = \mathbf{F}_4\mathbf{x}$$

$$X[0] = 5 + 5 + 5 + 5 = 20$$
$$X[1] = 5(1 - i - 1 + i) = 0$$
$$X[2] = 5(1 - 1 + 1 - 1) = 0$$
$$X[3] = 5(1 + i - 1 - i) = 0$$

So,

$$\mathbf{X} = \begin{bmatrix} 20 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Interpretation

- Only **DC component exists**
- No oscillations
- Signal is perfectly constant

This example clearly shows **what DC means physically**.

AMRITA VISHWA VIDYAPEETHAM DEEMED TO BE UNIVERSITY
Dr. Anirban Tarafdar
Assistant Professor, School of AI
Amrita Vishwa Vidyapeetham, Faridabad